

Multipolentwicklung

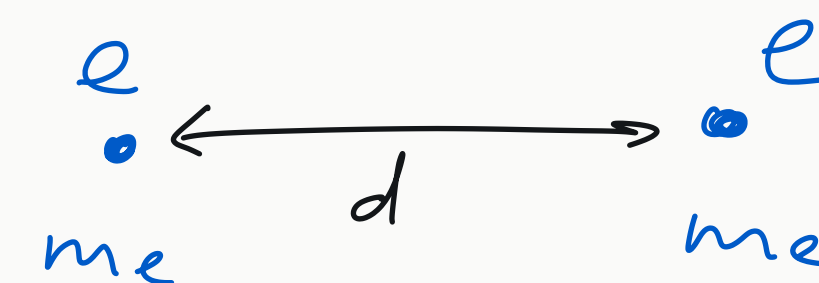
Die Coul. Wechselwirk. ist sehr stark



kleine Abweichung von der Neutralität



Dipol + Quadrupol Annäherung



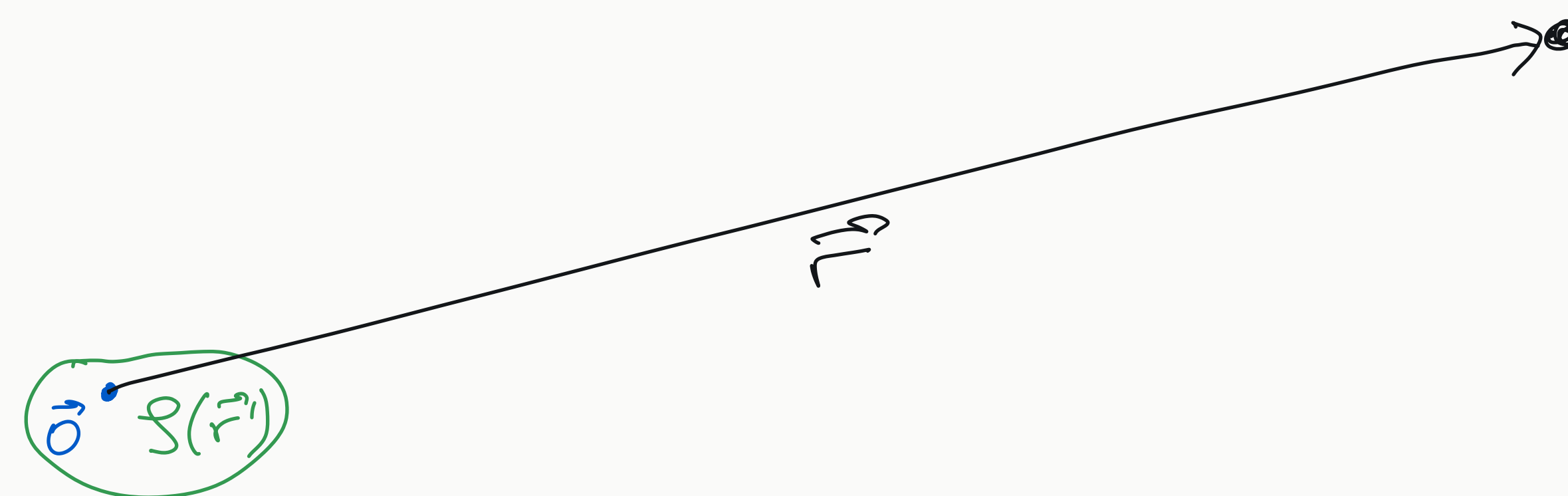
$$F_{\text{Coul}} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{d^2}$$

$$F_{\text{Grav}} = G \frac{m_e^2}{d^2}$$

$$\frac{F_{\text{Coul}}}{F_{\text{Grav}}} = \frac{\frac{1}{4\pi\epsilon_0} e^2}{G m_e^2} \sim 10^{42}$$

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3r'$$

Die Quelle ist sehr weit



Charact. Länge der $\rho(\vec{r}')$: $\ell \ll |\vec{r} - \vec{r}'| \sim r$



Das Objekt ist sehr weit

$$f = \frac{1}{|\vec{r} - \vec{r}'|}$$

$$f(\vec{r}') \approx f(\vec{0}) + \frac{\partial f}{\partial r'_\alpha} \bigg|_{\vec{r}'=\vec{0}} r'_\alpha + \frac{1}{2} \frac{\partial^2 f}{\partial r'_\alpha \partial r'_\beta} \bigg|_{\vec{r}'=\vec{0}} r'_\alpha r'_\beta + \dots$$

Einstein'sche Summen

$\alpha, \beta = 1, 2, 3 = x, y, z$
z.B. $r_2 = y$

$$f(\vec{r}') = \frac{1}{|\vec{r} - \vec{r}'|} = \frac{1}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}}$$

$$f(\vec{0}) = f(\vec{r}') \Big|_{\vec{r}' = \vec{0}} = \frac{1}{r}$$

$$\frac{\partial f}{\partial \vec{r}'} = \begin{pmatrix} \frac{\partial f}{\partial x'} \\ \frac{\partial f}{\partial y'} \\ \frac{\partial f}{\partial z'} \end{pmatrix} \quad \left| \quad \begin{aligned} \frac{\partial f}{\partial x'} &= \cancel{\frac{1}{2}} \frac{\cancel{2}(x-x')\cancel{(-1)}}{[(x-x')^2 + (y-y')^2 + (z-z')^2]^{3/2}} = \frac{x-x'}{|\vec{r} - \vec{r}'|^3} \\ \frac{\partial f}{\partial y'} &= \frac{y-y'}{|\vec{r} - \vec{r}'|^3} \\ \frac{\partial f}{\partial z'} &= \frac{z-z'}{|\vec{r} - \vec{r}'|^3} \end{aligned} \right.$$

$$\frac{\partial f}{\partial \vec{r}'} = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}$$

$$\frac{\partial f}{\partial \vec{r}'} \Big|_{\vec{r}' = \vec{0}} = \frac{\vec{r}}{r^3}$$

$$\frac{\partial f}{\partial x'} = \frac{x-x'}{|\vec{r}-\vec{r}'|^3} \leftarrow \frac{1}{((x-x')^2 + (y-y')^2 + (z-z')^2)^{3/2}}$$

$$\frac{\partial^2 f}{\partial x'^2} = \frac{\partial}{\partial x'} \left(\frac{\partial f}{\partial x'} \right) = -\frac{1}{|\vec{r}-\vec{r}'|^3} + (x-x') \left(-\frac{3}{2} \right) \frac{2(x-x')(-1)}{|\vec{r}-\vec{r}'|^5} = -\frac{1}{|\vec{r}-\vec{r}'|^3} + 3 \frac{(x-x')^2}{|\vec{r}-\vec{r}'|^5}$$

$$\frac{\partial^2 f}{\partial y' \partial x'} = \frac{\partial}{\partial y'} \left(\frac{\partial f}{\partial x'} \right) = (x-x') \left(-\frac{3}{2} \right) \frac{2(y-y')(-1)}{|\vec{r}-\vec{r}'|^5} = \frac{3(x-x')(y-y')}{|\vec{r}-\vec{r}'|^5}$$

$$\frac{\partial^2 f}{\partial r'_\alpha \partial r'_\beta} = \frac{-(\vec{r}-\vec{r}')^2 \delta_{\alpha\beta} + 3(r_\alpha - r'_\alpha)(r_\beta - r'_\beta)}{|\vec{r}-\vec{r}'|^5} \quad (***)$$

$$\left. \frac{\partial^2 f}{\partial r'_\alpha \partial r'_\beta} \right|_{\vec{r}'=\vec{0}} = \frac{-r^2 \delta_{\alpha\beta} + 3 r_\alpha r_\beta}{r^5}$$

$$\Phi(\vec{r}) \approx \frac{1}{4\pi\epsilon_0} \int d^3r' \rho(\vec{r}') \left[\frac{1}{r} + \frac{\vec{r}}{r^3} \cdot \vec{r}' + \frac{1}{2} \frac{-r^2 \delta_{\alpha\beta} + 3 r_\alpha r_\beta}{r^5} r'_\alpha r'_\beta \right] =$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} \underbrace{\int \rho(\vec{r}') d^3r'}_q + \frac{\vec{r}}{r^3} \cdot \underbrace{\int \rho(\vec{r}') \vec{r}' d^3r'}_{\vec{p}} + \frac{1}{2} \frac{-r^2 \delta_{\alpha\beta} + 3 r_\alpha r_\beta}{r^5} \underbrace{\int \rho(\vec{r}') r'_\alpha r'_\beta d^3r'}_{\textcircled{A}} \right]$$

$$r^2 \delta_{\alpha\beta} r'_\alpha r'_\beta = r^2 r'^2 = r_\alpha r_\beta \delta_{\alpha\beta} r'^2$$

Dipolmoment

$Q_{\alpha\beta} \leftarrow$ Quadrupoltensor

$$\textcircled{A} = -\frac{1}{2} \frac{r_\alpha r_\beta}{r^5} \int \rho(\vec{r}') r'^2 \delta_{\alpha\beta} + \frac{1}{2} \frac{r_\alpha r_\beta}{r^5} \int \rho(\vec{r}') 3 r'_\alpha r'_\beta d^3r' = \frac{1}{2} \frac{r_\alpha r_\beta}{r^5} \int \rho(\vec{r}') [3 r'_\alpha r'_\beta - r'^2 \delta_{\alpha\beta}] d^3r'$$

$$\Phi(\vec{r}) \approx \frac{1}{4\pi\epsilon_0} \int d^3r' \rho(\vec{r}') \left[\frac{1}{r} + \frac{\vec{r}}{r^3} \cdot \vec{r}' + \frac{1}{2} \frac{-r^2 \delta_{\alpha\beta} + 3 r_\alpha r_\beta}{r^5} r'_\alpha r'_\beta \right] =$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r} \underbrace{\int \rho(\vec{r}') d^3r'}_q + \frac{\vec{r}}{r^3} \cdot \underbrace{\int \rho(\vec{r}') \vec{r}' d^3r'}_{\vec{p}} + \frac{1}{2} \underbrace{\frac{-r^2 \delta_{\alpha\beta} + 3 r_\alpha r_\beta}{r^5} \int \rho(\vec{r}') r'_\alpha r'_\beta d^3r'}_{\textcircled{A}} \right]$$

$$r^2 \delta_{\alpha\beta} \frac{r'_\alpha r'_\beta}{r^2} = r^2 r'^2 = r_\alpha r_\beta \delta_{\alpha\beta} r'^2$$

Dipolmoment

$Q_{\alpha\beta} \leftarrow$ Quadrupoltensor

$$\textcircled{A} = -\frac{1}{2} \frac{r_\alpha r_\beta}{r^5} \int \rho(\vec{r}') r'^2 \delta_{\alpha\beta} + \frac{1}{2} \frac{r_\alpha r_\beta}{r^5} \int \rho(\vec{r}') 3 r'_\alpha r'_\beta d^3r' = \frac{1}{2} \frac{r_\alpha r_\beta}{r^5} \int \rho(\vec{r}') [3 r'_\alpha r'_\beta - r'^2 \delta_{\alpha\beta}] d^3r'$$

$$\Phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r} + \frac{\vec{r} \cdot \vec{p}}{r^3} + \frac{1}{2} \frac{r_\alpha r_\beta}{r^5} Q_{\alpha\beta} + \dots \right]$$

$\oplus$

$\begin{pmatrix} + \\ - \end{pmatrix}$

$\begin{pmatrix} + & & \\ - & + & - \end{pmatrix}$

Dipolfeld

$$\Phi_{\text{dip}} = \frac{1}{4\pi\epsilon_0} \frac{\vec{r} \cdot \vec{p}}{r^3}$$

$$(\ast\ast\ast) \rightarrow \left(\frac{\partial}{\partial r'} \left(\frac{\vec{r} \cdot \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right) = - \frac{\partial}{\partial r'} \left(\frac{\vec{r} \cdot \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right) \right) \Big|_{\vec{r}'=0} = \left[- \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \right] =$$

$$= \frac{-r^2 \delta_{\alpha\beta} + 3 r_\alpha r_\beta}{r^5}$$

Dipolfeld

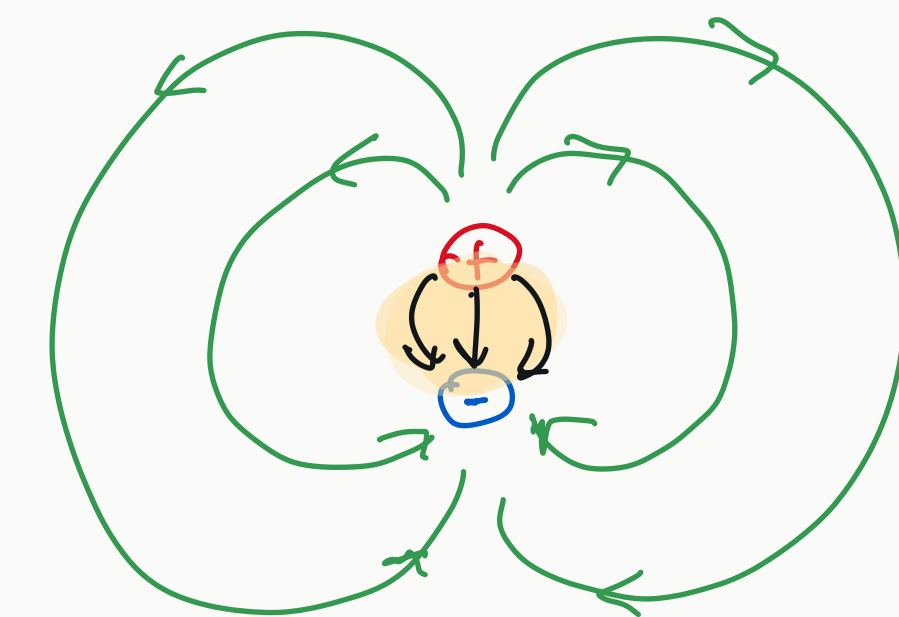
$$\vec{E}(\vec{r}) = - \frac{\partial \Phi_{\text{dip}}}{\partial \vec{r}} = - \frac{1}{4\pi\epsilon_0} \vec{p} \cdot \frac{\partial}{\partial \vec{r}} \left(\frac{1}{r^3} \right) = \frac{1}{4\pi\epsilon_0} \frac{-r^2 \overbrace{p_\alpha \delta_{\alpha\beta}}^{p_\beta} + 3 \overbrace{p_\alpha r_\alpha}^{(\vec{p} \cdot \vec{r})} r_\beta}{r^5} =$$

$$= \frac{1}{4\pi\epsilon_0} \left(- \frac{1}{r^3} \vec{p} + \frac{3(\vec{r} \cdot \vec{p})}{r^5} \vec{r} \right) = \frac{1}{4\pi\epsilon_0} \frac{3(\vec{r} \cdot \vec{p}) \vec{r} - r^2 \vec{p}}{r^5}$$

Die vollständige Formel für das Dipolfeld:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left[\frac{3(\vec{r} \cdot \vec{p}) \vec{r} - r^2 \vec{p}}{r^5} - \frac{4\pi}{3} \vec{p} \cdot \delta(\vec{r}) \right]$$

↓
nächste
Vorl.



Zum Weiterlesen:

Jackson 4.1, 4.2